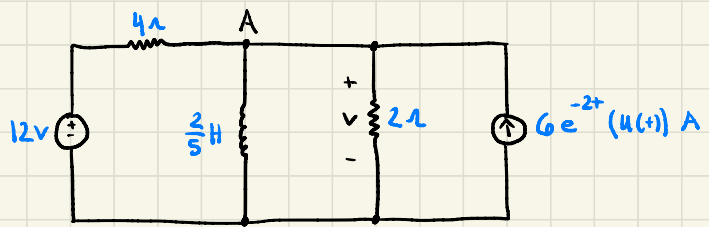
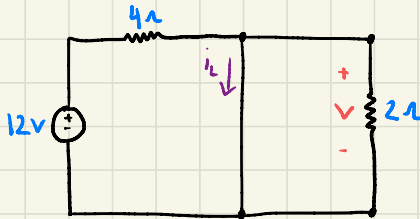


1.)

Find $v(t)$ for $t > 0$ Before $t = 0$

- The current source is "gone".
- The inductor acts as a short.

Initial current through inductor

$$-12 + 4i = 0$$

$$i = 3A$$

After $t > 0$

Node Voltage @ A:

$$4 \cdot \left(\frac{A - 12}{4} + i_L + \frac{A}{2} - 6e^{-2t} u(t) = 0 \right) \cdot 4$$

$$A - 12 + 4i_L + 2A - 24e^{-2t} u(t) = 0$$

$$3A + 4i_L = 24e^{-2t} + 12$$

$$3 \left(\frac{2}{5} \frac{di_L}{dt} \right) + 4i_L = 24e^{-2t} + 12$$

$$\frac{di}{dt} + \frac{10}{3}i = 20e^{-2t} + 10$$

$$e^{\frac{10}{3}t} \cdot i = \int e^{\frac{10}{3}t} \cdot 20e^{-2t} + 10$$

$$i = 15e^{-2t} + 3 \cdot 15e^{-\frac{10}{3}t}$$

$$v(t) = \frac{2}{5} \left(\frac{d}{dt} 15e^{-2t} + 3 - 30e^{-\frac{10}{3}t} \right) = 20e^{-\frac{10}{3}t} - 12e^{-2t}$$

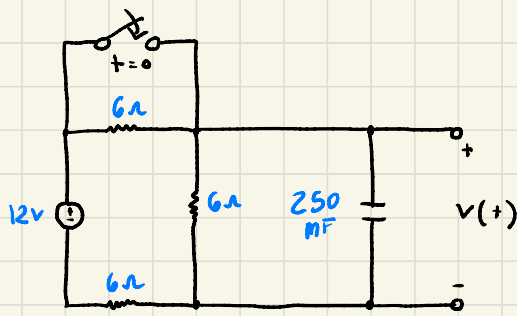
$$A = L \frac{di}{dt}$$

$$i_0 = 3A$$

$$C = -15$$

$$\int \frac{10}{3} dt = e^{\frac{10}{3}t}$$

2)



Find $v(t)$ for $t > 0$

Before $t > 0$

$V_c = V_{6\Omega}$ in Parallel

$$V_{6\Omega} = \frac{6}{18} = \frac{1}{3} \cdot 12V = 4V \quad \text{initial Voltage}$$

After $t > 0$

$V_c = V_{6\Omega}$ in Parallel

$$V_{6\Omega} = \frac{6}{12} = \frac{1}{2} \cdot 12 = 6V \quad \text{long term Voltage}$$

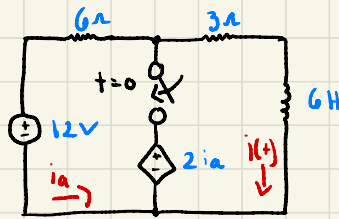
$$\text{Formula: } V(t) = V_{t>0} + (V_0 - V_{t>0}) e^{-t/\tau}$$

$$\tau = \frac{RC}{\downarrow} = 1.33s$$

$3\Omega \quad \downarrow \quad 250mF$

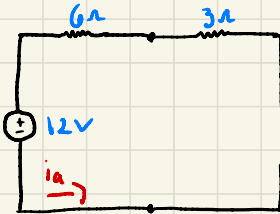
$$\begin{aligned} V(t) &= 6 + (4 - 6) e^{-1.33t} \\ &= 6 - 2e^{-1.33t} \end{aligned}$$

3.)



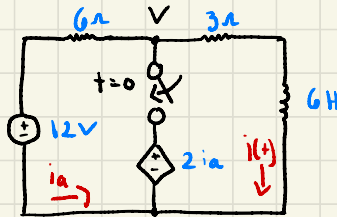
Find $i(t)$ for $t > 0$

Before $t > 0$



$$\text{Current: } \frac{12}{9} = \frac{4}{3} \text{ A}$$

After $t > 0$



$$\frac{2i_a - 12}{6} \Rightarrow i_a = -3 \text{ A}$$

Solve for τ

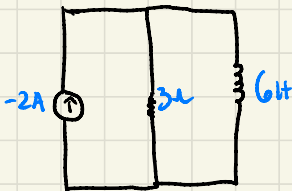
$$\tau = \frac{L}{R}$$

$$R_T = \frac{1}{6} + \frac{1}{3} = \frac{1}{2} \Omega \quad v = 2(-3) = -6 \text{ V}$$

$$i_L = \frac{-6}{3} = -2 \text{ A}$$

$$\text{Formula: } i_{t>0} + (i_0 - i_{t>0})e^{-t/\tau}$$

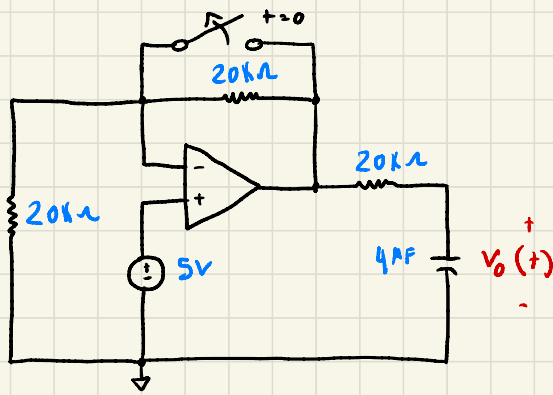
$$\frac{4}{3} + \frac{6}{3} = \frac{10}{3}$$



$$\tau = \frac{6}{3} = 2$$

$$i(t) = -2 + \left(\frac{4}{3} + 2\right)e^{-t/2} = -2 + \frac{10}{3}e^{-t/2}$$

4.)

Find $V_o(t)$ for $t > 0$

$$\text{Formula: } v(t) = V_{t>0} + (V_0 - V_{t>0}) e^{-t/\tau}$$

Before $t > 0$

$$V_0 = 5V \quad \text{since output resistance} = 0$$

After $t > 0$

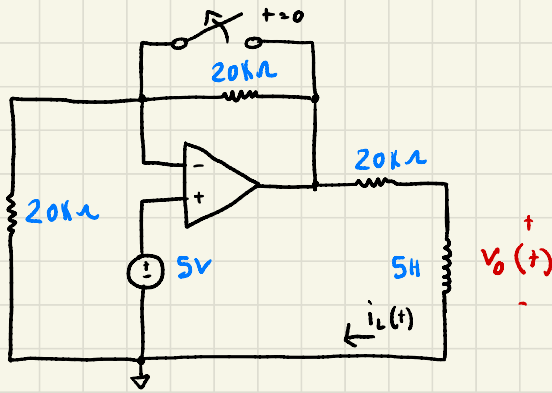
$$i_{Rf} = \frac{5}{20k} = 0.25 \text{ mA}$$

$$V_0 = V_0 - 5 = 5 = 10V$$

$$\tau = RC = 20k \cdot 4\mu F = 80 = \frac{1}{20} = 12.5$$

$$\begin{aligned} V_o(t) &= 10 + (5 - 10) e^{-12.5t} \\ &= 10 - 5 e^{-12.5t} \end{aligned}$$

5)

Find $V_o(t)$ for $t > 0$ Before $t > 0$

$$V_o = 5V$$

$$i_L = \frac{5}{20k} = 0.25 \text{ mA}$$

After $t > 0$

$$V_o = 10V$$

$$i_L = \frac{10}{20k} = 0.500 \text{ mA}$$

$$\tau = \frac{L}{R} = \frac{5H}{20k} = \frac{1}{4000} = \frac{1}{4000} = 4000$$

$$i(t) = 0.500 + (0.25 - 0.500)e^{-4000t}$$

$$V_o(t) = 5 \left(\frac{d}{dt} 0.500 - 0.25e^{-4000t} \right)$$

$$= 5e^{-4000t}$$